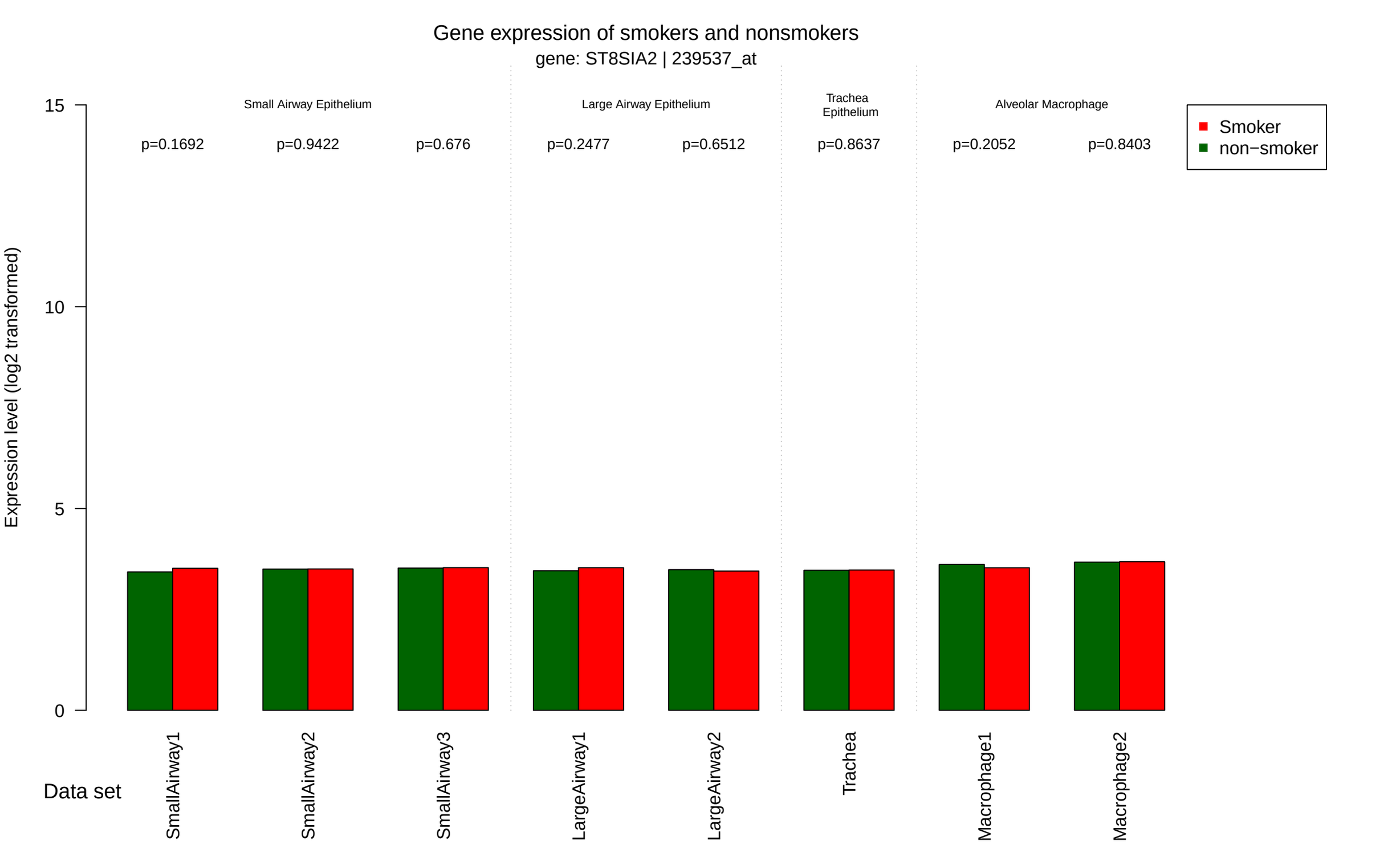
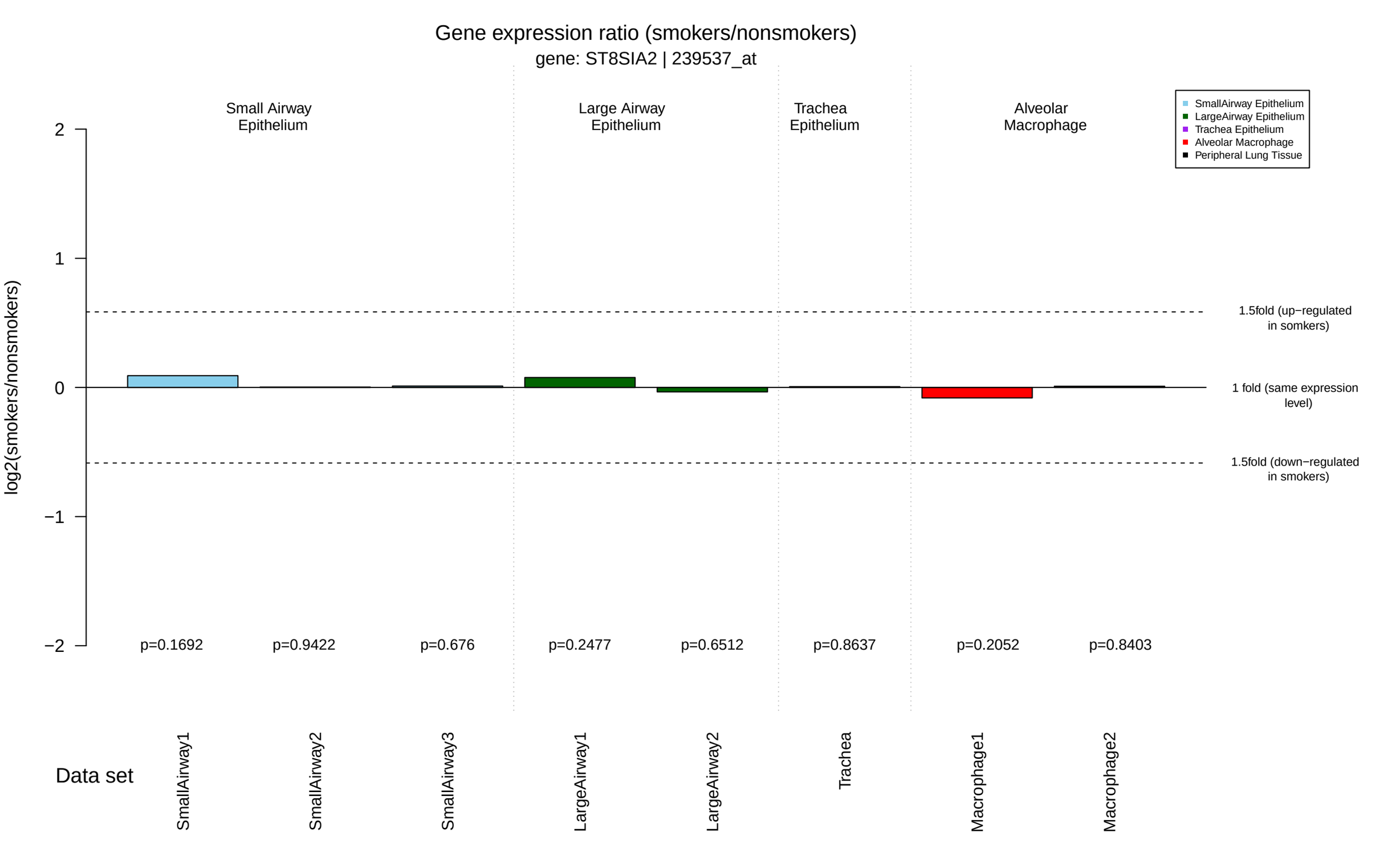
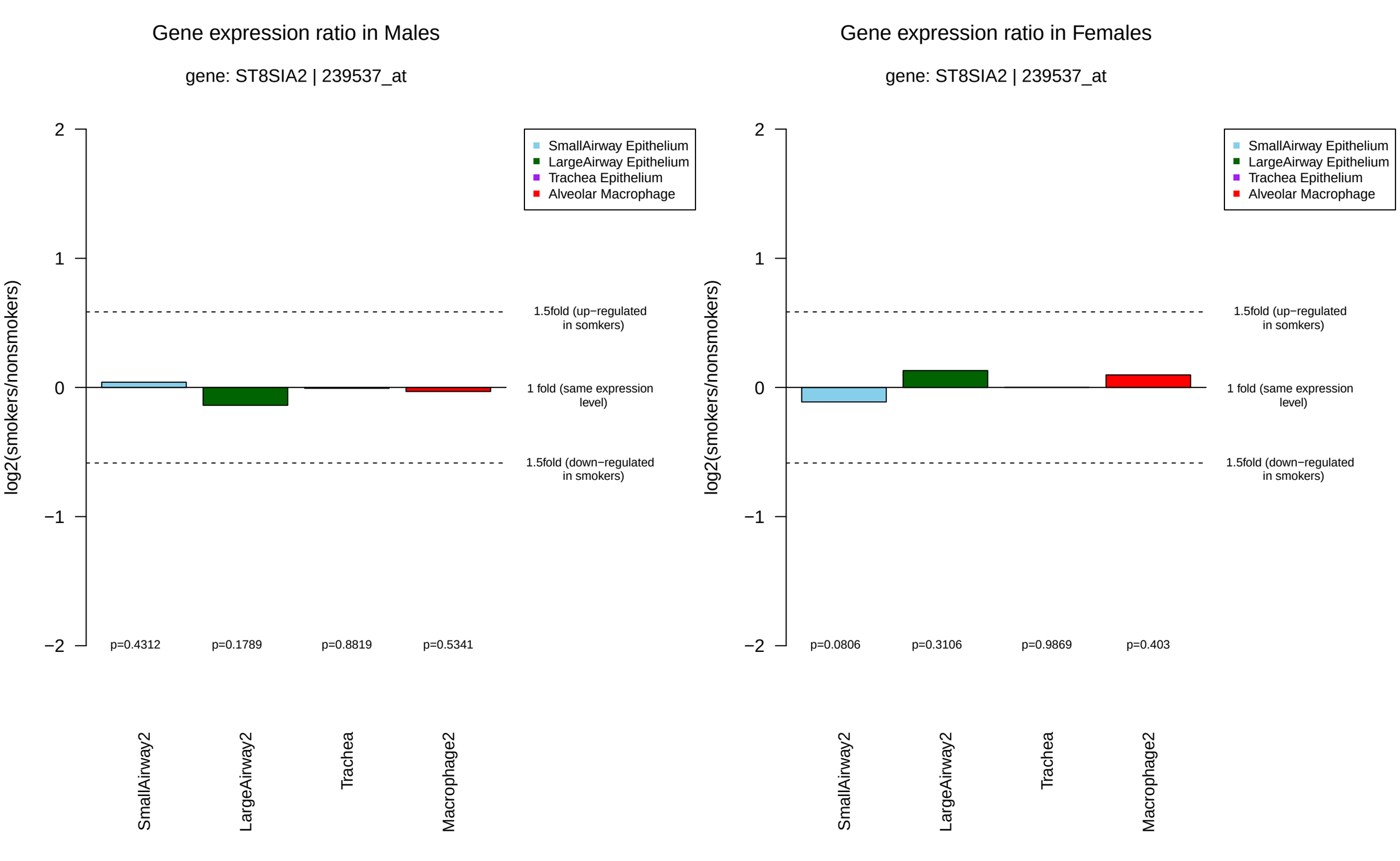
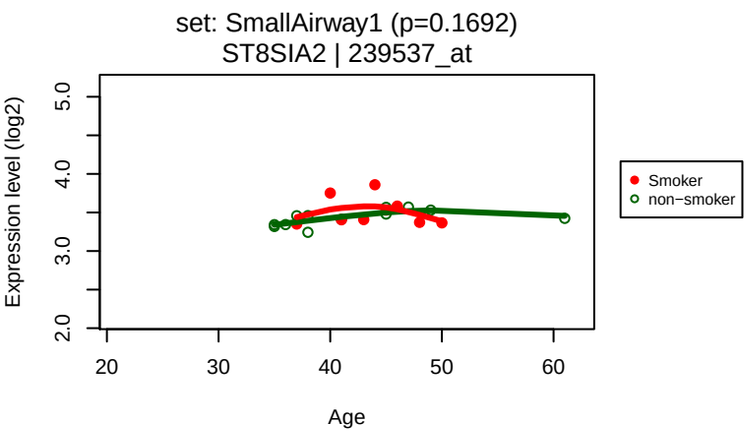
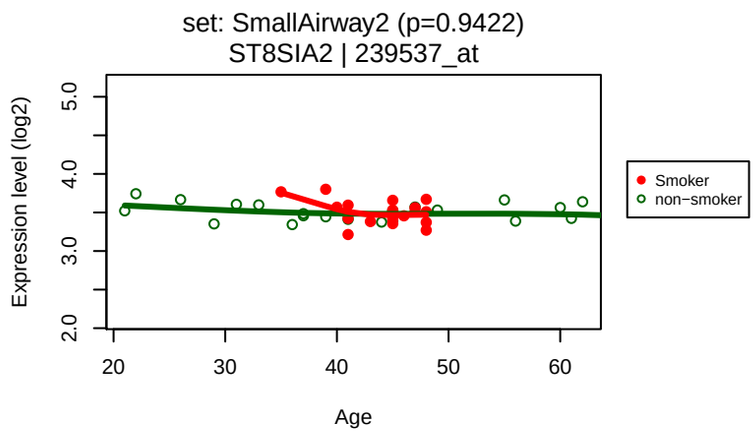
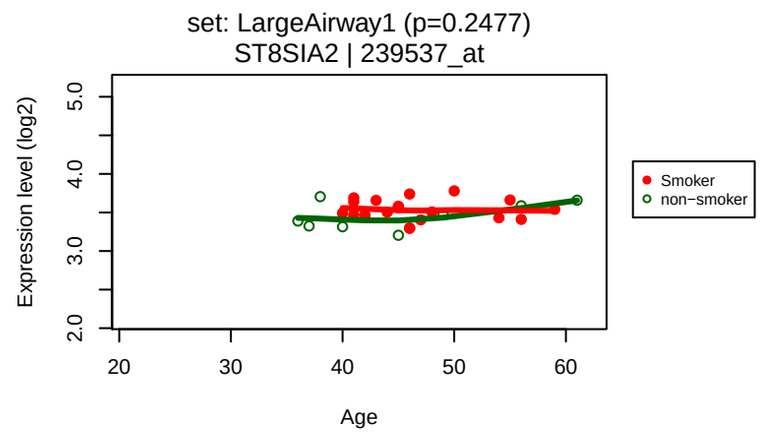

Gene expression of smokers and nonsmokers
gene: ST8SIA2 | 239537_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1692
p=0.9422
p=0.676
p=0.2477
p=0.6512
p=0.8637
p=0.2052
p=0.8403
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

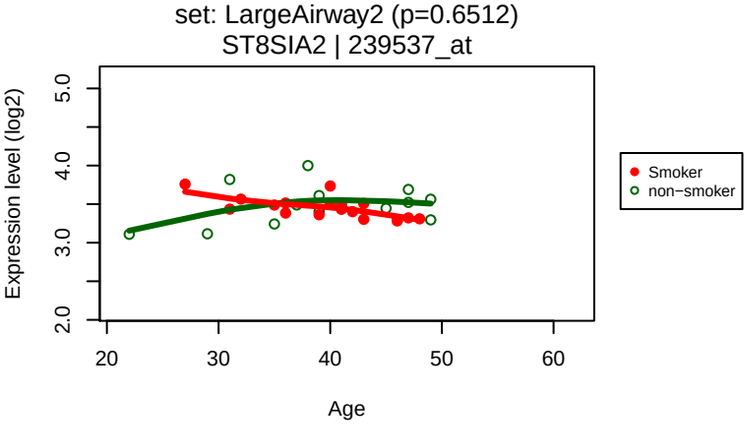

Gene expression ratio (smokers/nonsmokers)
gene: ST8SIA2 | 239537_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1692
p=0.9422
p=0.676
p=0.2477
p=0.6512
p=0.8637
p=0.2052
p=0.8403
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

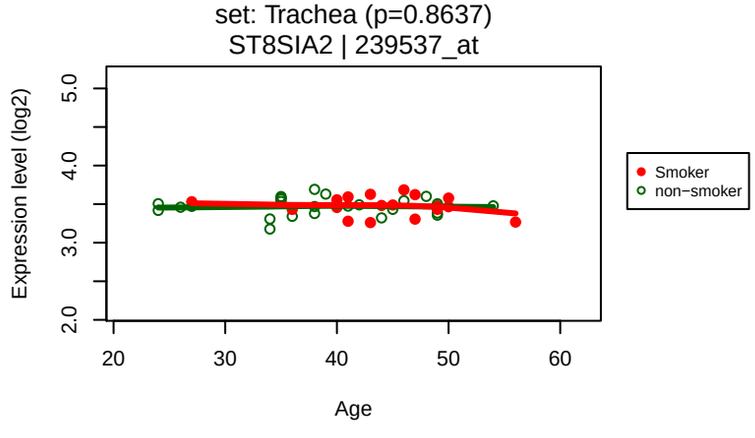

Gene expression ratio in Males
gene: ST8SIA2 | 239537_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4312
p=0.1789
p=0.8819
p=0.5341
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ST8SIA2 | 239537_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0806
p=0.3106
p=0.9869
p=0.403
SmallAirway2
LargeAirway2
Trachea
Macrophage2

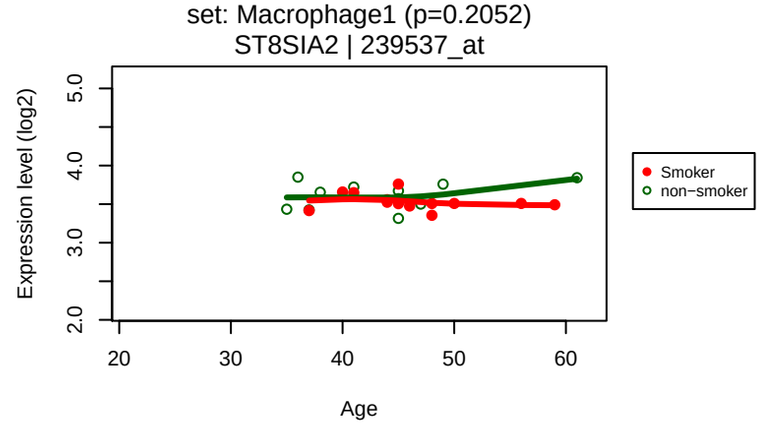

set: SmallAirway1 (p=0.1692)
ST8SIA2 | 239537_at
Expression level (log2)
Age
Smoker
non−smoker

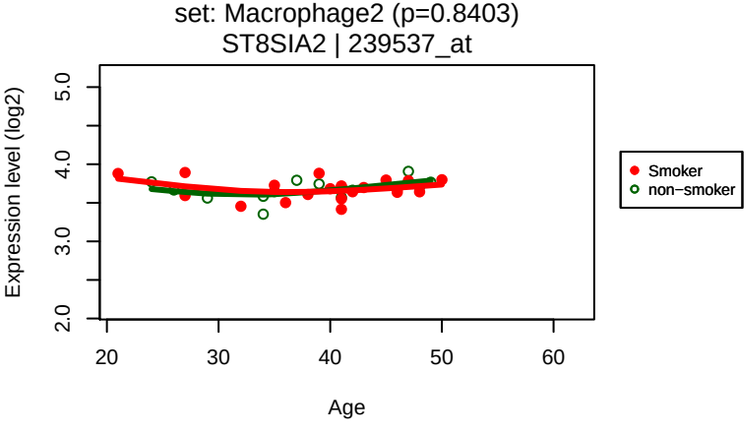

set: SmallAirway2 (p=0.9422)
ST8SIA2 | 239537_at
Expression level (log2)
Age
Smoker
non−smoker

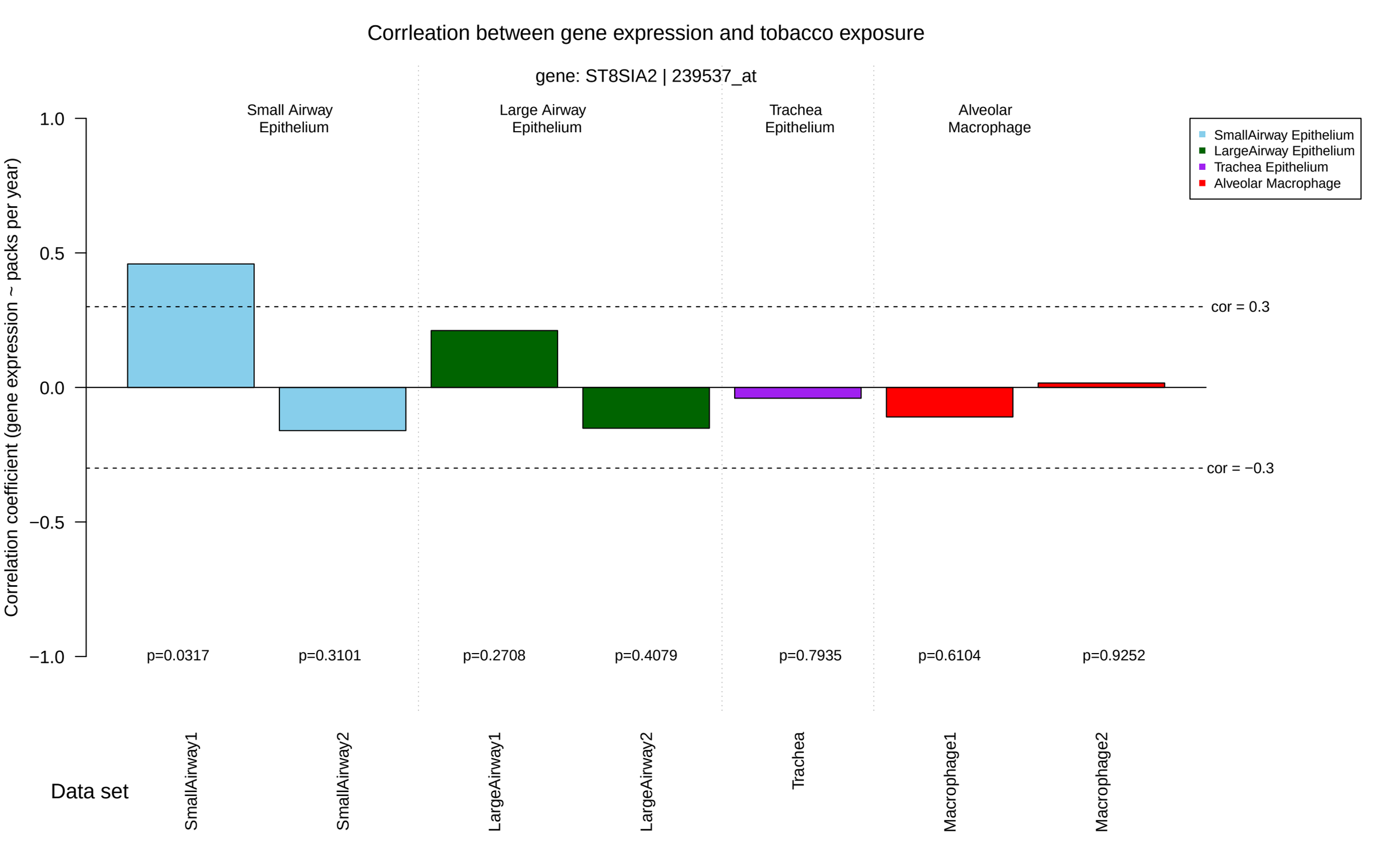

set: LargeAirway1 (p=0.2477)
ST8SIA2 | 239537_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6512)
ST8SIA2 | 239537_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.8637)
ST8SIA2 | 239537_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2052)
ST8SIA2 | 239537_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.8403)
ST8SIA2 | 239537_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ST8SIA2 | 239537_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0317
p=0.3101
p=0.2708
p=0.4079
p=0.7935
p=0.6104
p=0.9252
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2